

DSC 257R: UNSUPERVISED LEARNING

FALL 2025

Course Syllabus

Instructor Information

Faculty of Record: **Sanjoy Dasgupta**

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Instructional Assistant: **TBD**

Email: TBD@ucsd.edu

Office hours: **by appointment**

Zoom address: **TBD** (use during office hours only)

Course location & Zoom session meeting times

Canvas Course Location: <https://canvas.ucsd.edu/courses/28952>

Live Session Schedule **Wednesdays at 6pm**

Course Description

DSC 257R presents a broad view of unsupervised learning. Topics to be covered include: descriptive statistics; clustering; projection, singular value decomposition, and spectral embedding; common probability distributions; density estimation; graphical models and latent variable modeling; sparse coding and dictionary learning; autoencoders, shallow and deep; and self-supervised learning. The focus throughout will be on understanding the modeling assumptions behind different methods, their statistical and algorithmic characteristics, and common issues that arise in practice.

Students will apply techniques from this course in a variety of mathematical exercises as well as in programming assignments using Jupyter notebooks.

Prerequisites

This course assumes that students have prior experience with:

- Basics of probability and statistics
- Basics of linear algebra
- Basics of multivariate calculus
- Machine learning at the level of DSC 255R
- Programming in Python and Jupyter notebooks

Some of the required mathematical background will be reviewed as it is needed in the course.

Textbooks

There is no required textbook for the course. Students seeking supplementary reading material are encouraged to consult:

Kevin Murphy. Probabilistic Machine Learning: An Introduction. MIT Press.

Evaluation and Grading

The course material is divided into modules, each with an accompanying worksheet containing conceptual questions, mathematical exercises, and programming projects. The weekly homework will consist of the worksheets for that week's modules. All assignments will be turned in on Gradescope.

The final grades will be based entirely on submitted work, with the following weighting.

Assignments	% of Grade
Weekly Homework	100%
Total	100%

Grading Scale

This course uses the following grading scale.

Letter Grade	Percentage
A	94% - 100%
A-	90% - 93.99%
B+	87% - 89.99%
B	84% - 86.99%
B-	80% - 83.99%
C+	77% - 79.99%
C	74% - 76.99%
C-	70% - 73.99%
D+	67% - 69.99%
D	64% - 66.99%
D-	60% - 63.99%
F	< 60%

Course Workflow

Every week, students should:

- Download the slides for that week
- Watch the videos for that week and fill in missing details on the slides while doing so

- Attend the online discussion section
- Attend office hours and/or ask questions on the Q&A platform
- Complete the worksheets for each of the week's modules and submit on Gradescope

UCSD Policies

Code of Conduct

All participants in the course are bound by the **University of California Code of Conduct** (<https://aisc.uci.edu/students/index.php>)

Netiquette

Be respectful. Be sensitive. Be aware. Effective written communication and open academic dialogue are crucial for sustaining a learning community that is respectful, considerate, relevant, creative, and thought-provoking. In an online classroom, expressions, meaning, and tone can quickly be taken out of context, making it imperative that online learners adhere to the communication guidelines below:

- Treat your classmates with respect.
- Be thoughtful and open in a discussion.
- Be aware and sensitive to different perspectives.
- Build one another up and encourage one another to succeed.

The following behavior should be avoided:

- Using insulting, condescending, or abusive words.
- Using all capital letters, which comes across as SHOUTING.
- Contacting learners or posting advertisements and solicitations.
- Posting copyrighted material.

Academic Integrity

Academic Integrity is expected of everyone at UC San Diego. This means you must be honest, fair, responsible, respectful, and trustworthy in your actions.

Lying, cheating, or other forms of dishonesty will not be tolerated because they undermine learning and the University's ability to certify students' knowledge and abilities. Thus, any attempt to get, or help another get, a grade by cheating, lying, or dishonesty will be reported to the Academic Integrity Office and result in sanctions. Sanctions can include an F in the class and suspension or dismissal from the University.

So, think carefully before you act. Before you act, ask yourself the following questions: a) is my action honest, fair, respectful, responsible, and trustworthy, and b) is my action authorized by the instructor? If you are unsure, don't ask a friend; ask your instructor, instructional assistant, or the Academic Integrity Office. You can learn more about academic integrity at academicintegrity.ucsd.edu.

Disability Services

University of California, San Diego is committed to providing a barrier free environment for persons with documented disabilities. If you have a disability and feel you need accommodations in this course, please contact the Disability Services Center, 858-534-4382, or apply for services online at <https://osd.ucsd.edu/>. DSC approved accommodations will be provided for students who present a Faculty Notification Letter from the DSC.